

StyleCart – Fashion E- Commerce Database

Chapter – 2

2.0.Introduction:

Entity Analysis is a fundamental step in database design that identifies the important objects involved in a system and defines their attributes, keys, and relationships. In the StyleCart Fashion E-Commerce system, the main entities identified are Customer, Brand, Category, Product, Cart, Cart_Item, Order, Payment, and Review. The Customer entity stores customer information, while the Brand and Category entities manage the classification of fashion products. The Product entity maintains details about available fashion items, including brand, category, price, size, and stock information. The Cart and Cart_Item entities manage products selected by customers, and the Order and Payment entities handle order processing and transaction details. The Review entity stores customer feedback and ratings for products. This entity analysis provides a clear structure for designing the StyleCart database and helps in maintaining data consistency, integrity, and efficient data management.

2.1.Entity: Customer

Attribute	Key Type
Customer_Id	Primary Key
First_Name	Not Null
Last_Name	Not Null
Email	Unique Key
Phone	Unique Key
Password	Not Null
Address	Not Null
City	-
State	-
Pincode	-

Entity: Brand

Attribute	Key Type
Brand_Id	Primary Key
Brand_Name	Unique Key
Description	-
Country	-

Entity: Category

Attribute	Key Type
Category_ID	Primary Key
Category_Name	Unique Key
Description	-

Entity: Product

Attribute	Key Type
Product_ID	Primary Key
Product_Name	Not Null
Brand_ID	Foreign Key
Category_ID	Foreign Key
Price	-
Size	-
Color	-
Material	-
Stock	-
Product_Image	-

Entity: Cart

Attribute	Key Type
Cart_ID	Primary Key
Customer_ID	Foreign Key
Grand_Total	-
Total_Items	-

Entity: Cart_Item

Attribute	Key Type
CartItem_ID	Primary Key
Cart_ID	Foreign Key
Product_ID	Foreign Key
Quantity	-
Total_Price	-

Entity: Order

Attribute	Key Type
Order_ID	Primary Key
Customer_ID	Foreign Key
Order_Date	-
Total_Amount	-
Order_Status	-

Shipping_Address	-
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Entity: Payment

Attribute	Key Type
Payment_ID	Primary Key
Order_ID	Foreign Key
Payment_Method	-
Payment_Date	-
Payment_Status	-
Amount_Paid	-

Entity: Review

Attribute	Key Type
Review_ID	Primary Key
Customer_ID	Foreign Key
Product_ID	Foreign Key
Rating	-
Comment	-
Review_Date	-

2.2.Entity Relationships

Entity 1	Relation	Entity 2	Cardinality	Participation
Customer	Owns	Cart	One-to-One	Total / Total
Customer	Places	Order	One-to-One	Partial / Total
Customer	Writes	Review	One-to-One	Partial / Partial
Brand	Manufactures	Product	One-to-One	Partial / Total
Category	Contains	Product	One-to-One	Partial / Total
Cart	Contains	Cart_Item	One-to-One	Total / Total
Cart_Item	Includes	Product	Many-to-Many	Total / Total
Order	Contains	Order_Item	One-to-One	Total / Total
Order_Item	Refers To	Product	Many-to-Many	Total / Total
Order	Has	Payment	One-to-One	Total / Total
Order	Has	Shipment	One-to-One	Total / Partial
Product	Receives	Review	One-to-One	Partial / Partial

2.3.Conclusion

The StyleCart Fashion E-Commerce Database Management System is designed to efficiently manage all major activities of an online fashion store. The database stores and organizes customer information, product details, brands, categories, shopping carts, orders, payments, and reviews in a structured manner. Using entities, relationships, primary keys, and foreign keys ensures data integrity, minimizes redundancy, and improves data retrieval.

This database design provides a reliable foundation for managing online shopping operations efficiently and can be further enhanced by adding advanced features such as inventory management, order tracking, discounts, and personalized product recommendations.